

Lab Experiment-05

5. Write a Python function to analyze the histogram of the given input image based on color levels using Open CV.

Code:

```
!pip install opencv-python matplotlib
import cv2
import matplotlib.pyplot as plt
from google.colab import files
uploaded = files.upload()
filename = list(uploaded.keys())[0]
img = cv2.imread(filename)
def analyze_histogram(image):
    colors = ('b', 'g', 'r') # Blue, Green, Red channels
    plt.figure()
    plt.title("Color Histogram")
    plt.xlabel("Pixel Intensity")
    plt.ylabel("Frequency")
    for i, col in enumerate(colors):
        hist = cv2.calcHist([image], [i], None, [256], [0, 256])
        plt.plot(hist, color=col)
        plt.xlim([0, 256])
    plt.show()
if img is None:
    print("Error: Image not loaded")
else:
    analyze_histogram(img)
```

Output:

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packag

image.jpg

image.jpg(image/jpeg) - 22746 bytes, last modified: 4/30/2026 - 100% done

Saving image.jpg to image (4).jpg

